

Fernando Fernández Andueza

Gameplay and AI Programmer

andueza013.github.io/PortfolioGameDev anduezadev@gmail.com linkedin.com/in/fernando-fernandez-andueza anduezadev.itch.io
Valencia, Spain. EU citizen, open to relocation within Spain and to remote work. Spanish (native), English (intermediate).

Profile

I built all four enemy archetypes for Beerserker: The Stolen Brew in StateTree, with C++ and Blueprints, and wrote XEMA, my own C++ and OpenGL engine, with an ECS and deferred PBR rendering. I work with Unreal and Unity, and before games I spent two and a half years as a fullstack developer on ERP software.

Technical skills

Engines & AI: Unreal Engine 5 (UE5) with C++ & Blueprints, Unity (C#), StateTree, Behavior Trees, EQS (Environment Query System), AI Perception, Animation systems, Object pooling

Programming languages: C++ (C++17, C++20, C++23, STL), C#, C, ARM64 assembly, Lua

Graphics & systems: OpenGL 4.5 DSA, Deferred shading / PBR, Vertex & fragment shaders, Entity Component System (ECS), Multithreading & job systems, Memory management & pointers

Tools: Visual Studio debugger, Unreal Insights, RenderDoc, Git, Perforce, Plastic SCM

Outside games: TypeScript / JavaScript, HTML / CSS, Vue.js, SQL Server, Java, VB.NET

Experience

Gameplay Programmer, Oniric Tales

Dec 2024 – Jun 2025

Roadtrip: The Engine of Madness (Unity)

- Designed and built core gameplay systems: a component-based dialogue system, an AI perception system integrated with the behaviour trees, and a radio communication system.
- Also worked on the Behavior Graph, general gameplay mechanics, profiling and bug fixing.

Fullstack Developer, AHORA

Sep 2020 – Mar 2023

- Enterprise management system: frontend with Vue.js, TypeScript and DevExtreme; backend in VB.NET and C# on SQL Server.
- Built products, components and extensions with FlexyGo, an in-house low-code framework.
- Handled client incidents, often on site. Worked in Scrum.

Selected projects

Beerserker: The Stolen Brew

Unreal Engine 5, C++

Final-year project at ESAT, team of about 20. Developed by Grumpy Games and ESAT, published on Steam by ESAT (Early Access, free).

- Implemented the four enemy archetypes (melee, ranged, explosive and a tank mini-boss) with StateTree in C++ and Blueprints, with custom tasks and conditions.
- Built an object pool as a plugin, reusable in other projects, for the many enemies, particles, decals and projectiles.
- Used EQS to pick where the ranged enemy retreats to, based on distance to the player and line of sight. The tank favours the melee attacks it has used least, so it doesn't keep repeating the same one.
- Built the enemy spawners, set up and uploaded the Steam builds, and took part in designing the architecture of a custom action system based on GAS (Gameplay Ability System).

XEMA (custom C++ engine)

C++, OpenGL 4.5

- ECS, deferred PBR, shadow mapping, instancing, a JobSystem for async mesh loading, Lua scripting and PhysX.
- Ported to Nintendo Switch, tested only on an emulator. PS4 port in progress on a devkit: the ECS and simple-geometry rendering work; mesh loading, textures, instancing, lights, shadows and PBR are still missing.

Abandoned Hospital

Unreal Engine 5, C++

- Class exercise with AI Perception and EQS on a behaviour tree; I later rebuilt it on my own, with the AI rewritten in StateTree.

Education & certifications

HND in Video Game Programming, ESAT (Escuela Superior de Arte y Tecnología), Valencia

2023–2026

Specialising in C++: my own engine from scratch, procedural generation (Wave Function Collapse, cellular automata, L-systems), a templated FSM, A* pathfinding and optimization in assembly to find and fix bottlenecks in inner loops.

DAM (Multiplatform Application Development), IES El Grao, Valencia

2018–2020

Java for desktop and Android, relational databases, HTML and CSS, OOP fundamentals and version control.

Certified Scrum Master & Product Owner, Scrum.org / International Agile Institutes